

benefit from using symbolic solvers; **CoT appears to be a poor (but universal) approximation to such solvers**. When possible, LLMs should be paired with symbolic solvers at inference time when solving symbolic tasks to achieve consistently better performance over direct answer **and** CoT.

6 DISCUSSION AND RELATED WORK

Where is CoT helping and why? Our results showing CoT improvement for math and logic aligns well with early work on CoT for LLMs such as Scratchpads (Nye et al., 2022). As CoT gained popularity, its application has broadened to tasks that canonically do not require multiple steps. It can often yield small improvements over direct answering. We believe this led to the current prevailing sentiment that deliberation should improve performance on any task requiring some type of reasoning (our original claim from Section 2). However, our results show a clear separation between performance on non-symbolic and symbolic tasks. If, in theory, any question could benefit from deliberation, why is CoT only benefiting the questions that can be solved through symbolic manipulation? Our results from Section 5 suggest that the primary benefit of CoT comes in the ability to execute symbolic steps and track their output. Not all tasks have this feature: for example, questions from CommonsenseQA can hardly be translated into formally grounded and executable solution plans. Datasets like StrategyQA may feature multiple steps of reasoning, but executing those steps is not complex, so the benefits of CoT are small. It is unclear whether explicitly instilling models with particular modes of deliberation, like process of elimination for multiple choice questions, might make them more effective for non-symbolic tasks, or whether there’s a fundamental limitation imposed by their pre-training data. We leave this distinction for future work.

Can we improve CoT further? Our work treats chain-of-thought variants that explicitly don’t involve multiple inferences. There is evidence that using additional calls to LLMs can help (Du et al., 2023; Yao et al., 2023; Besta et al., 2023; Chen et al., 2024), but these methods use significantly increased computation, and careful benchmarking sometimes reveals that naive techniques are as good as iterative ones (Olausson et al., 2024). However, past theoretical results show that Transformers are augmented in a fundamental way by CoT (Liu et al., 2023b; Merrill & Sabharwal, 2024); we believe this indicates the potential for improving CoT beyond prompt-based CoT. On the other hand, recent methods showing benefit from “internalizing” CoT (Deng et al., 2024) may indicate that explicit generation of intermediate tokens is not used to its full potential.

Limitations One set of tasks we do not cover in our experiments (except for BiGGen Bench) is long-horizon planning. However, many works in the literature have already discussed the efficacy of planning with CoT. We also do not address the data contamination of some of these models on the datasets. We try to mitigate this by including multiple models, datasets (new and old), and our meta-analysis. For more discussion of planning and dataset contamination, see Appendix I.

7 CONCLUSION

In this work, we characterize the performance of prompt-based CoT through a meta-analysis of the literature and experiments across different models, datasets, and prompts. We find that CoT predominantly helps on math and formal logic, largely due to its ability to trace the intermediate steps of a problem. But CoT rarely outperforms tool-augmented approaches for these same problems. We believe that CoT remains a powerful technique, but to give improvement across a wider range of NLP tasks, research should move beyond prompt-based CoT to new paradigms like search, interacting agents, or better fine-tuned models.

REPRODUCIBILITY

For our experiments, we provide in-depth details of how we evaluated models on each dataset in Section 4.1 and Appendix C. Furthermore, we release all prompts for every dataset on Huggingface, including per model output and sampling parameters. For our meta-analysis of the literature, we describe our filtering criteria and process of annotating experiments into high-level categories in Section 3 and Appendix B. We also release the full list of papers in our meta-analysis together with extracted experimental comparisons and task category annotations.

ACKNOWLEDGMENTS

We acknowledge George Tsoukalas for providing insightful feedback throughout the project. We also thank Kaj Bostrom and Eunsol Choi for reviewing and providing feedback on drafts of the work. This work was partially supported by NSF CAREER Award IIS-2145280 (to Durrett), NSF CAREER Award 2339729 (to Mahowald), the NSF AI Institute for Foundations of Machine Learning (IFML), the Sloan Foundation via a Sloan Research Fellowship, and a grant from Open Philanthropy.

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